

SCX Paper Framework Review Report — Five-Reviewer Synthesis

Review Date: 2026-06-26 Review Method: 5 Independent AI Reviewers in Parallel + Synthesis Adjudication
Reviewer Roles: Narrative Arc / Theory Construction / Evidence Chain / Competition Positioning / Cross-Domain Extension

1. Five-Reviewer Consensus Matrix

Judgment Dimension	Narrative	Theory	Evidence	Competition	Cross-Domain	Consensus
EGP P1 Should Be Published First						
SCX-Theory standalone submission is high-risk						
V(s) multiplicative decomposition is the weakest theoretical link				—		—
“Estimated 29–48%” needs retraining verification					—	
Should merge Theory + MLIP into one paper					—	—
MedMNIST/CIFAR experiments currently harmful				—	—	
“Six-dimension coverage” should become a single core narrative					—	—
Arrow analogy should be removed				—		—
Relationship with Data Shapley should be complementary				—	—	—
Narrow domains from 7 to 1–2				—	—	—

2. Core Fear Diagnosis: “Others Will Think I’m a Fraud”

2.1 Is the Fear Reasonable?

Reasonable. Not because SCX is fraudulent, but because the narrative order determines the reviewer’s first impression.

2.2 Three Attack Paths

Attack Path 1: Triviality Attack (“Isn’t this just conditional probability?”)

- SCX’s three Propositions: Prop 1 is mathematically almost trivial
- Reviewer: “Conditional risk is a concept from Kolmogorov (1933). What’s new here?”
- **Defense:** Prove the cost of ignoring state-conditionality in real material systems—EGP P1’s $9.7\times$ per-batch RMSE difference

Attack Path 2: Patchwork Attack (“Clustering + model selection + active learning pieced together with a new name”)

- Reviewer: “State discovery = clustering. Expert reliability = per-cluster accuracy. Data value = heuristic product.”
- **Defense:** Two-layer descriptors (ErrorDrivenEncoder) are the core—the $2.3\times$ improvement from F1 $0.253 \rightarrow 0.585$ proves this is not trivial clustering

Attack Path 3: Castle in the Air Attack (“No real experimental verification”)

- Reviewer: “You estimate 29–48% improvement but never actually re-trained.”
- **Defense:** A single retraining run (4–8 hours compute) can plug this gap—this is currently the most fatal gap

3. Optimal Paper Framework: “Triple Jump”

3.1 Overall Strategy

EGP P1 (Independent, concrete) $\rightarrow$ SCX-MLIP (Theory+Application merged, core battle)

Core logic: **First concrete to establish a base $\rightarrow$ then abstract to build theory (but not standalone submission) $\rightarrow$ then cross-domain extension**

3.2 First Jump: EGP Paper 1

Dimension	Description
Core Contribution	Gauge-fixed shared+correction ACE parameterization
Target Journal	npj Computational Materials / Physical Review Materials / JCTC
Audience	Materials/computational chemistry community
Length	4–6 pages main text + SI
Data	AlN v3: 425–534 frames DFT already trained; GaN/AlN v4 supercomputing in progress

Section Structure

1. Introduction

- Gauge ambiguity problem in multi-element ACE potentials
- Limitations of existing methods (post-hoc splicing of single ACE training)
- Contribution: shared+correction parameterization + coefficient-level gauge fixing

2. Methods

2.1 Shared+Correction ACE Architecture

$$E = \sum_i [cB_i + c_{\{Z_i\}}B_i + b_{\{Z_i\}}]$$

2.2 Gauge Freedom and Post-Hoc Projection

$$c_Z \rightarrow c_Z - g, c \rightarrow c + g \quad (g = \sum_Z c_Z)$$

2.3 Why Soft Constraint Fails (gradient competition analysis)

2.4 Training Protocol

3. Results: AlN v3

3.1 EOS (V, B vs DFT)

3.2 Elastic Constants (C-C)

3.3 Phonon Forces

3.4 Gauge Violation: $8.77 \rightarrow 4.6 \times 10^1$

4. Discussion

- Significance of gauge fixing for potential merging (AlN+GaN→AlGaN)
- For pure AlN, gauge fixing is formal completeness
- Future work: state-conditioned expert reliability (one sentence)

Red Lines (Do Not Cross)

- Do not insert SCX theory (at most one sentence of future work in Discussion)
- Do not include SCX noise detection Figs (Fig 4–8 reserved for SCX-MLIP)
- Do not claim Model B is a “new runtime MoE potential architecture”
- Do not claim transferability to AlGaN (unless experimental data exists)

3.3 Second Jump: SCX-MLIP (Theory+Application Merged, Core Battlefield)

	<u>Dimension</u>	<u>Description</u>
Core Contribution	State-conditioned data quality framework + MLIP verification	
Target Journal	npj Computational Materials / Nature Communications	
Audience	MLIP community + data valuation community	
Length	Main text 1/3 theory (condensed) + 2/3 experiments	

Section Structure

1. Introduction

- MLIP data is expensive, but data value is not fixed
- Same potential performs vastly differently across configuration regions (cite)
- Core problem: How to assess data quality and expert reliability in a state-co

2. Theory: State-Conditioned Expertise (condensed to ~3 pages main text)

2.1 Core Definitions

$R_m(s)$, $SCX_m(s)$, four-category data classification

2.2 Three Propositions (condensed version)

Prop 1 (restructured): Regret lower bound for global ranking

Prop 2 (restructured): Suboptimality of error-driven sampling under noise

Prop 3: Advantage of state-conditioned weighting

2.3 Non-Identifiability Theorem

Without additional assumptions, noise and inherent difficulty cannot be dis

→ Multi-expert information is necessary

2.4 State Discovery: Two-Layer Architecture

L1: Domain knowledge encoder → L2: ErrorDrivenEncoder (error-correlated sub

3. Methods: SCX Pipeline for MLIP
 - 3.1 ACE Descriptor State Encoder
 - 3.2 Two-Layer State Discovery
 - 3.3 State-Level Data Classification
 - 3.4 Action Policy
4. Experiments
 - 4.1 AlN v3: Two-Layer State Discovery (core)
 - One-layer vs two-layer comparison (F1: 0.253 → 0.585, Top-3: 37.8% → 94.6)
 - Noise distribution (100% in thermal + MLMD)
 - Phonon redundancy (50--80% compressible)
 - 4.2 AlN v3: Retraining Validation PO mandatory
 - SCX denoising → retrain → measured force RMSE vs baseline
 - vs random drop / loss-based drop
 - Training fmax vs test error: r=0.966
 - 4.3 Cross-Material Validation
 - SCX analysis on public Si/Cu/MgO datasets
 - 4.4 Synthetic Experiments
 - Theoretical boundary verification (numerical demonstration of non-identifi
5. Discussion
 - SCX is a data diagnosis tool, complementary to model architecture improvement
 - Extension potential: engineering simulation, medical data
 - Limitations: cold start requires initial labels; sensitivity to state partiti

Content Not Included in the Paper

Deleted Item	Reason
V(s) multiplicative decomposition	Heuristic definition; replace with four-category rules
MedMNIST / CIFAR experiments	Currently too weak, harms credibility
“Six-dimension coverage” narrative	Replace with “calibrate→route→audit” three-layer structure

Deleted Item	Reason
Arrow impossibility theorem analogy	Technical error + excessive packaging
LLM / Music / Semi / Vision / Sim domain mentions	Narrow down to MLIP
“Data Poisoning Defense” concept	Replace with “Data Quality Audit”
“General framework” claim	Replace with “methodology for MLIP data quality with cross-domain potential”

3.4 Third Jump: SCX-Sim or SCX-Health (Pick One, Dive Deep)

Option	Target Journal	Prerequisites	Priority
SCX-Sim	Nature Computational Science	Requires industry partner data	Preferred if partner exists
SCX-Health	npj Digital Medicine	MedMNIST experiments need GPU upgrade and valid results	Choose this if no partner

Recommend choosing one, not both. Diving deep into one domain is 10× more effective than broad coverage.

4. Revised Paper Sequence

Order	Paper	Core Problem	Target Journal	Data Status
1	EGP: Gauge- fixed shared+correction ACE	ACE potential gauge fixing and merging	npj CompMat / PRM	AIN v3 trained, GaN/AIN v4 supercomput- ing
2	SCX- MLIP: State- Conditioned Data Quality for MLIP	State- conditioned data quality framework + MLIP verification (theory 1/3 + experiments 2/3)	npj CompMat / NatComm	Needs retraining verification + public datasets
3	SCX- Sim or SCX- Health	Cross-domain extension	NatCompSci / npj DigMed	Needs partner data or GPU experiments

Changes: Original SCX-Theory (standalone submission) and SCX-MLIP merged into one paper. SCX-Sim and SCX-Health: pick one.

5. Mandatory Experiment Checklist Before Submission

Priority	Experiment	Compute Cost	Consequence of Absence
P0	AlN v3 post-denoising retrain: Remove fmax>5 frames or SCX Top-2 noise states → retrain ACE → measure force RMSE	4–8h CPU	Reviewer rejects outright—“estimation is not experiment”
P1	Public MLIP dataset cross-validation: Download public Si/Cu/MgO training data → run SCX two-layer analysis → report noise/redundancy findings	1–2d analysis (zero DFT)	“Only works on AlN” criticism cannot be answered

Priority	Experiment	Compute Cost	Consequence of Absence
P1	Synthetic experiments: Generate synthetic data with known noise structure → verify SCX recovers state structure → verify non-identifiability theorem boundaries	0.5d	Theory papers lack controlled experiments
P2	Comparison with random drop / loss-based drop: Add control groups in denoising experiment → show SCX-guided denoising outperforms naive drop	Included in P0	“Anyone can remove high-error frames” criticism cannot be answered

Priority	Experiment	Compute Cost	Consequence of Absence
P3	CIFAR-10 noise detection GPU upgrade: ResNet-18 + 50 epoch + 5 seeds	1–2d GPU	Needed if making cross-domain claim; not needed if narrowed to MLIP

P0 cannot be skipped. This experiment determines whether the paper can upgrade from “interesting observation” to “verified contribution.”

6. Theory Reinforcement Checklist

Priority	Modification	Explanation
P0	Remove Arrow impossibility theorem analogy	Technical error, excessive packaging
P0	Restructure Prop 1	Change from “no global optimum exists” to “regret lower bound for global ranking is $\Omega(p \cdot \delta_{\min})$ ”

Priority	Modification	Explanation
P1	Add non-identifiability theorem	Explicit proof that without additional assumptions, noise and inherent difficulty are indistinguishable $\rightarrow$ necessity of multi-expert information
P1	Add cold-start protocol	Two-stage: Phase 1 unlabeled proxy estimation $\rightarrow$ Phase 2 acquisition
P2	Replace $V(s)$ with four-category rules	Avoid heuristic multiplicative decomposition controversy
P2	Abstract state discovery	In theory paper, treat state mapping as oracle; ErrorDrivenEncoder as Appendix instantiation

7. Competition Positioning Improvements

Competitor	Current Positioning	Improved Positioning
Data Shapley	Replacement relationship (SCX $V(s)$ vs Data Shapley φ_i)	Complementary relationship: SCX coarsely estimates states $\rightarrow$ Data Shapley fine-tunes within selected states

Competitor	Current Positioning	Improved Positioning
MoE (Shazeer/DeepMD)	Two-stage vs end-to-end	Scenario difference: Heterogeneous experts + incremental extension + auditable → SCX; homogeneous end-to-end → traditional MoE
LLM Data Governance	Not the main battlefield	One sentence in Introduction + immediate pivot: Mention LLM scenario as motivation, then shift to scientific simulation (which has gold labels)
Conditional Risk	New concept	Perspective difference: Discretized states ≠ continuous conditional risk; discretization is prerequisite for auditability

8. Inter-Paper Narrative Relationships

EGP Paper 1 (concrete, materials method)

"We discovered in this work that expert merging requires gauge fixing + state
→ Provides empirical motivation for SCX-MLIP

SCX-MLIP (theory + application, merged version)

"EGP P1's empirical observations → formalized as state-conditioned expertise
Theory 1/3 length + Experiments 2/3 length

SCX-Sim or SCX-Health (cross-domain)

"Applicability of SCX framework in a completely different domain"

Citation Relationships

EGP P1 (does not cite SCX) Published independently

SCX-MLIP (cites EGP P1 as motivation) Submitted after EGP publication

SCX-Sim/Health (cites SCX-MLIP as methodological foundation) Submitted after SCX-M

9. One-Sentence Summary

The core of “one strike and it’s done” is not how elegant the theory is, but that by the time the reviewer finishes page three, they are already convinced by AlN’s $9.7\times$ per-batch difference and the measured improvement from denoising retraining—at which point you tell them the theoretical framework behind it, and they can only think “this theory makes sense, because it explains the experimental phenomena I just saw.”

This report was generated by 5 independent AI reviewers reviewing in parallel, followed by synthesis. Individual reviewer reports are located in the `CodeKnowledge/agent_outputs/` directory.